

Learned Bounded Off-Grid Refinement for Sparse Tensor-OMP in mmWave MIMO-OFDM ISAC

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Abstract—Sparse tensor recovery is a natural model for mmWave MIMO-OFDM integrated sensing and communication, but dictionary quantization limits physically interpretable parameter estimates. We study learned, bounded interpolation between Tensor-OMP grid supports and deterministic local-refinement candidates. The local three-dimensional implementation learns one scalar and improves measurement-domain NMSE over grid Tensor-OMP by 5.8956 dB on average across $L \in \{2, 4, 8\}$ in a five-seed, 450-scene synthetic study (seed-mean 95% CI [5.7830, 6.0083] dB). A separate 210-parameter feature controller for two-dimensional CDL measurements predicts delay and angle interpolation scales and is trained with permutation-invariant, alias-aware physical supervision. On frozen held-out CDL-A/C evaluation it yields gains of 6.2286 dB in channel NMSE, 10.1241 ns in delay RMSE, and 0.5277° in projected-angle RMSE. Zero-shot CDL-D degrades channel and delay estimates, bounding the profile-generalization claim. A matched dense-tensor comparison further shows that the NOMP-inspired path is the accuracy-oriented point whereas the learned scalar is a lower-compute amortized alternative. A separate deterministic repeated-refinement diagnostic tests a restricted high-load computation-quality trade-off but does not support an SNR-robust early-stop claim. These results support a compute-accuracy and external-robustness view of bounded post-support refinement, not accuracy supremacy or a fully integrated deep-unfolded network.

Index Terms—Integrated sensing and communication, model-driven learning, Tensor-OMP, off-grid estimation, MIMO-OFDM, CDL channel.

I. INTRODUCTION

Millimeter-wave MIMO-OFDM systems offer the bandwidth and aperture needed for high-resolution sensing while simultaneously supporting communication links. In integrated sensing and communication (ISAC), the same received waveform can carry information about both propagation channels and physical targets. This shared geometry is attractive, but it also makes estimation harder: channel taps, target angles, delays, and Doppler shifts must be recovered from a high-dimensional observation while remaining interpretable enough for both communication and sensing tasks.

A useful abstraction is to view the problem as sparse recovery over an angle-delay-Doppler tensor. Dominant paths or targets occupy only a small number of tensor atoms, so Tensor-OMP provides a natural model-driven baseline. A purely grid-based estimator, however, loses accuracy when physical parameters fall between dictionary bins. This off-grid mismatch is especially important when the recovered support should remain physically meaningful rather than merely reduce a black-box loss.

This work studies learned bounded refinement after Tensor-OMP support selection. The central operation interpolates

between a coarse grid support and a deterministic local candidate, then recomputes the least-squares reconstruction. Two learned implementations are evaluated rather than conflated: a one-parameter scalar interpolator for the local three-dimensional experiment and a separate 210-parameter feature controller for delay-angle CDL measurements. The latter uses permutation-invariant, alias-aware physical supervision. The contribution is therefore a constrained post-support learning mechanism, not a completed multi-layer Tensor-OMP network.

The empirical story has four deliberately separate layers. Local synthetic data isolate scalar interpolation, yielding a mean 5.8956 dB gain for $L \in \{2, 4, 8\}$ over 450 held-out scenes. A matched multi-SNR comparison places the NOMP-inspired method at the accuracy-oriented end of the Pareto display and the learned scalar at a lower-compute, lower-accuracy point; this explicitly rules out an accuracy-supremacy claim. Scaled held-out CDL-A/C evaluation tests the distinct feature controller and improves channel NMSE by 6.2286 dB, delay RMSE by 10.1241 ns, and projected-angle RMSE by 0.5277° on average, while zero-shot CDL-D exposes the profile-shift boundary. Finally, a deterministic repeated-refinement study evaluates a plateau gate at high load and 20 dB; its SNR sweep prevents a broader adaptive-depth claim.

The paper makes five scoped contributions:

- 1) It formulates joint channel and target parameter estimation as sparse recovery over an angle-delay-Doppler tensor for mmWave MIMO-OFDM ISAC.
- 2) It introduces bounded post-support interpolation and instantiates it as a one-parameter local model and a separate feature-conditioned delay-angle controller with permutation-invariant supervision.
- 3) It provides controlled local evidence and scaled held-out CDL-A/C physical evaluation, together with a zero-shot CDL-D failure boundary.
- 4) It reports a matched NMSE-runtime Pareto comparison that discloses NOMP-inspired accuracy dominance in the matched synthetic regime and frames learning as compute amortization plus external robustness.
- 5) It characterizes a restricted deterministic high-load early-stop diagnostic and reports the SNR sensitivity that prevents a broader adaptive-depth claim.

The manuscript remains conservative. A unified multi-stage learned estimator, trained DeepMIMO physical evaluation, instrumented-receiver validation, robust hardware-impairment training, and full multi-target CRLB tightness remain open.

The matched NOMP-inspired result is restricted to known-order dense tensors from the sinusoidal generator, and the overall result is standards-aligned but not profile-universal.

II. RELATED WORK

a) Sparse and tensor-based estimation in ISAC.: High-frequency communication and sensing channels are often represented by a small number of dominant propagation or target components, which makes sparse recovery a natural baseline for mmWave MIMO-OFDM estimation. Matching pursuit and orthogonal matching pursuit established greedy sparse approximation as an interpretable alternative to dense inversion [1], [2], [3], while multiple-measurement-vector sparse recovery formalized the value of shared support across related observations [4]. These classical references explain why a Tensor-OMP scaffold is a natural starting point, but they do not by themselves address off-grid physical parameters or trainable target supervision.

Tensor formulations add another important prior: angle, delay, Doppler, antenna, subcarrier, and snapshot dimensions can be kept separable instead of being collapsed into a single vectorized dictionary. Tensor factorization and uniqueness results provide the broader mathematical background for such separable representations [5], [6], [7]. Recent ISAC neighbors include unified massive-MIMO tensor estimation [8], the scalable decoupled-and-fused PLAIN architecture [9], and structured angle–delay–Doppler CP decomposition for mixed near-/far-field AFDM sensing [10]. Deep-learning-aided CP-ALS further illustrates how learning can accelerate classical tensor decomposition [11]. Our implementation instead retains greedy Tensor-OMP support selection and learns only a bounded interpolation after a local candidate has been constructed.

Sparse mmWave channel-estimation and hybrid-precoding work provides a second classical neighborhood. Channel-estimation and sparse-precoding formulations for millimeter-wave MIMO [12], [13] and broader surveys [14], [15] motivate physically structured channel descriptions. The present manuscript adds held-out CDL-A/C evaluation while keeping unseen CDL-D as a failure boundary.

b) Deep unfolding for wireless and JCAS.: Deep unfolding converts iterative optimization or sparse-recovery procedures into trainable networks whose layers retain an algorithmic interpretation [16], [17]. Recent surveys consolidate wireless applications and design trade-offs [18], [19]. In joint communication and sensing, [20] unfolds a hybrid-beamforming optimizer, while learned trimmed-ridge recovery targets sparse mmWave channel estimation [21]. DDA-Net [22] unfolds ADMM for multi-snapshot channel reconstruction, uses a Doppler-domain denoiser and delay oversampling, and reports QuaDRiGa-to-CDL generalization. AirFM-DDA extends the same physical domain toward foundation-model channel tasks [23]. These are deeper trainable inference paths than the present post-support controller; our current evidence does not establish a comparable end-to-end unfolding network.

c) Off-grid sparse recovery and model-driven refinement.: Discretized sparse estimators are sensitive to grid

mismatch: a target or path between atoms can be represented by a biased or spread support. Off-grid sparse Bayesian inference [24] and its recent AFDM-ISAC specialization [25] treat physical offsets as inference variables. Newtonized OMP provides the closest non-learning mechanism: it alternates sparse support detection with continuous Newton refinements for sparse OFDM-ISAC target estimation and includes measurement validation [26]. Our learned interpolation is bounded and support-conditioned, but the present experiments do not provide a matched-accuracy comparison against Newtonized OMP or off-grid SBL.

d) Learning objectives for ISAC target estimation.: Learning-based ISAC estimators must balance communication-channel reconstruction with sensing-oriented target accuracy. OFDM radar-communication fusion and ISAC surveys establish the dual-functional signal-processing context [27], [28]; newer reviews organize deep-learning ISAC tasks and high-resolution parametric/neural sensing methods [29], [30], [31]. OFDM/OTFS comparisons show that waveform choice can materially affect joint performance [32]. A closer channel-estimation neighbor is [33], which uses radar-derived angle information inside a deep-unfolded sparse channel-estimation network.

The CDL controller operates in an FDMA virtual-array setting with an explicit multi-target parameter objective. Because the same target set can be listed in different orders, its training loss uses assignment-based matching to make the target-parameter term permutation-invariant. This use is algorithmic rather than a novelty claim: the Hungarian method and later assignment formulations are classical optimization tools [34], [35]. The manuscript also treats CRLB and impairment sensitivity as bounded analysis surfaces: standard estimation theory anchors the high-SNR CRLB check [36], while non-ideal massive-MIMO hardware analysis motivates keeping hardware impairment robustness as a future-work boundary rather than an established claim [37].

e) External benchmark context.: DeepMIMO provides a widely used ray-tracing benchmark context for mmWave and massive-MIMO research [38], [39]. Sionna supplies the standards-aligned link-level implementation used for the reported CDL evaluation [40], [41]; QuaDRiGa is an established spatially consistent channel simulator and the source-domain benchmark used by DDA-Net [42], [22]. The current DeepMIMO runs remain scalar-transfer proxies, so the trained physical external claim is carried by CDL rather than ray tracing.

f) Closest-work boundary.: Table I makes the closest-work boundary explicit. The contribution is the evidence-backed combination of greedy tensor support selection, learned bounded local interpolation, permutation-invariant multi-target supervision, and visible failure boundaries, rather than novelty in any one ingredient.

III. SYSTEM MODEL AND PROBLEM FORMULATION

We consider a mmWave MIMO-OFDM ISAC setting in which the received measurements are represented over a separable angle-delay-Doppler tensor dictionary. Let $\mathcal{Y}$ denote the

TABLE I

NEAREST-NEIGHBOR POSITIONING. “EXTERNAL EVIDENCE” RECORDS EACH WORK’S REPORTED VALIDATION SCOPE, NOT A HEAD-TO-HEAD RESULT.

Method		Core inference	Off-grid mechanism	Learned component	External evidence / boundary
NOMP-OFDM-ISAC [26]		greedy OMP target detection	continuous Newton refinement	none	simulated and semi-anechoic V2X sensing; range/velocity rather than learned 3-D tensor recovery
Unified ISAC [8]	tensor	classical tensor decomposition	parametric tensor factors	none	joint channel/target tensor anchor; no unfolded bounded-refinement controller
PLAIN [9]		decoupled subspace/CS estimation plus fusion	super-resolution estimators	none	scalable multi-dimensional architecture; no learned sparse-recovery path
DDA-Net [22]		ADMM unfolding in DDA domain	delay oversampling	Doppler denoiser	QuaDRiGa training and CDL-B zero-shot channel NMSE; not ISAC multi-target supervision
Learned OMP (ours)	Tensor-refinement	greedy support selection plus LS	bounded interpolation toward a deterministic local candidate	one scalar locally; separate 210-parameter CDL controller	matched NOMP-inspired accuracy/runtime comparison; held-out CDL-A/C physical metrics; zero-shot CDL-D failure retained

observed tensor and let the unknown scene be approximated by a sparse set of atoms indexed by angle, delay, and Doppler coordinates. The sparse-support view is useful because each selected atom has a physical interpretation, while the tensor representation avoids erasing separability across propagation and sensing dimensions.

For tensor dimensions $N_a \times N_\tau \times N_\nu$, the executable synthetic model is

$$\mathcal{Y} = \sum_{\ell=1}^L \beta_\ell \mathbf{a}_{N_a}(b_{a,\ell}) \circ \mathbf{a}_{N_\tau}(b_{\tau,\ell}) \circ \mathbf{a}_{N_\nu}(b_{\nu,\ell}) + \mathcal{W}, \quad (1)$$

where $\beta_\ell \in \mathbb{C}$ is the path coefficient, $\circ$ denotes the outer product, and

$$[\mathbf{a}_N(b)]_n = N^{-1/2} \exp(j2\pi nb/N), \quad n = 0, \dots, N-1. \quad (2)$$

Each continuous bin is $b_{d,\ell} = k_{d,\ell} + \delta_{d,\ell}$, with grid index $k_{d,\ell}$ and bounded offset $\delta_{d,\ell}$. Equations (1)–(2) are identical to the separable atoms used by the off-grid sample generator and least-squares reconstruction code.

The paper focuses on the FDMA virtual-array regime used by the current evaluation package. Within this scope, a target or path is represented by a coarse grid location plus a possible off-grid offset. The coarse grid gives a discrete support for Tensor-OMP, and the off-grid offset captures the local mismatch between physical parameters and dictionary bins. The estimator therefore has two linked jobs: it must select a plausible sparse support and then refine the selected atoms without drifting away from physically meaningful coordinates.

The main reconstruction metric is measurement-domain NMSE,

$$\text{NMSE}_{\text{dB}} = 10 \log_{10} \frac{\|\hat{\mathcal{Y}} - \mathcal{Y}_0\|_F^2}{\|\mathcal{Y}_0\|_F^2}, \quad (3)$$

where $\mathcal{Y}_0$ is the noise-free tensor. Target-parameter errors use the minimum-cost Hungarian assignment between estimated and reference target sets; per-axis RMSE is then computed on the matched pairs with circular bin distance. The local regime uses a $128 \times 16 \times 32$ tensor. The standards-aligned CDL regime uses a 64×4 frequency/ULA observation, CIR delay truth, and the identifiable projected broadside angle.

The controller is trained and validated on disjoint CDL-A/C seeds and frozen before held-out evaluation; CDL-D

is reserved for zero-shot profile-shift testing. The standards-aligned channel generator follows the CDL family in 3GPP TR 38.901 and is executed through Sionna [41], [40]. DeepMIMO O1/I3 scalar-transfer proxies are available but do not yet use the same trained physical delay/angle contract.

IV. LEARNED BOUNDED REFINEMENT

The method keeps Tensor-OMP support selection and least-squares reconstruction explicit. Learning controls only how far the estimate moves from a selected grid atom toward a deterministic local-search candidate. Figure 2 separates the two learned implementations from the repeated-refinement depth diagnostic; they share a bounded interpolation principle but are not one integrated network.

A. Tensor-OMP Support and Bounded Interpolation

Let $\mathbf{b}_{0,\ell}$ denote the coarse tensor-grid coordinates selected for component ℓ , and let $\mathbf{b}_{*,\ell}$ be the candidate returned by deterministic bounded local search around that support. The learned coordinates are

$$\hat{\mathbf{b}}_\ell = \mathbf{b}_{0,\ell} + \alpha \odot (\mathbf{b}_{*,\ell} - \mathbf{b}_{0,\ell}), \quad 0 < \alpha_d < 1.2, \quad (4)$$

where a scalar α is broadcast over axes in the local implementation and axis-specific scales are used in the CDL implementation. Given the design matrix $\mathbf{A}(\hat{\mathbf{B}})$ formed from all refined atoms, coefficients and reconstruction are recomputed as

$$\hat{\beta} = \arg \min_{\beta} \|\mathcal{Y} - \mathbf{A}(\hat{\mathbf{B}})\beta\|_2^2, \quad \hat{\mathcal{Y}} = \mathbf{A}(\hat{\mathbf{B}})\hat{\beta}. \quad (5)$$

Thus the learned output remains an inspectable physical support followed by an LS estimate.

B. Local Three-Dimensional Scalar Model

For the $128 \times 16 \times 32$ angle–delay–Doppler experiment, the trainable model contains one unconstrained scalar a . Its bounded interpolation scale is

$$\alpha = 1.2 \sigma(a), \quad (6)$$

and a is optimized through the differentiable LS reconstruction to minimize linear-scale measurement NMSE. The coarse support and local candidate are prepared before this optimization.

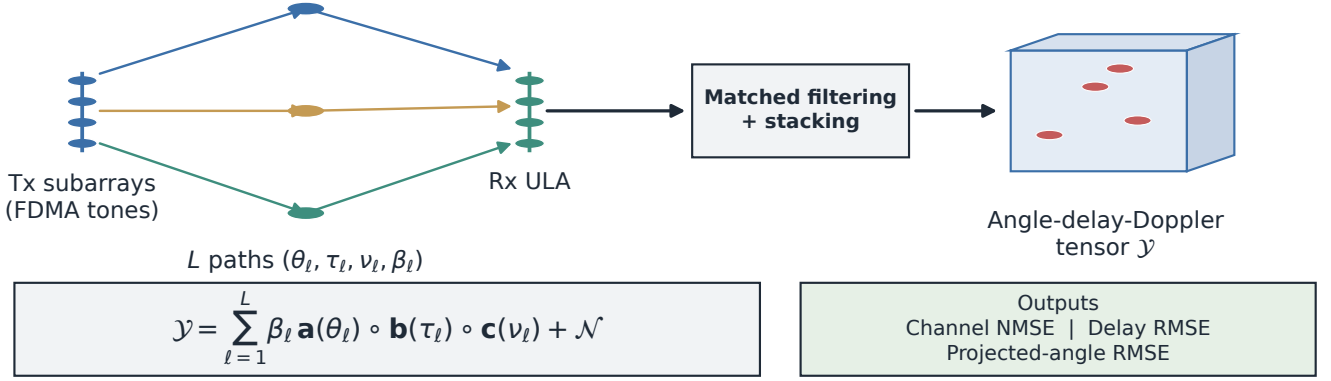


Fig. 1. FDMA MIMO-OFDM ISAC observation model. A small set of paths or targets couples the transmit subarrays and receive ULA. Matched filtering and stacking preserve a separable angle–delay–Doppler tensor whose sparse atoms retain physical meaning.

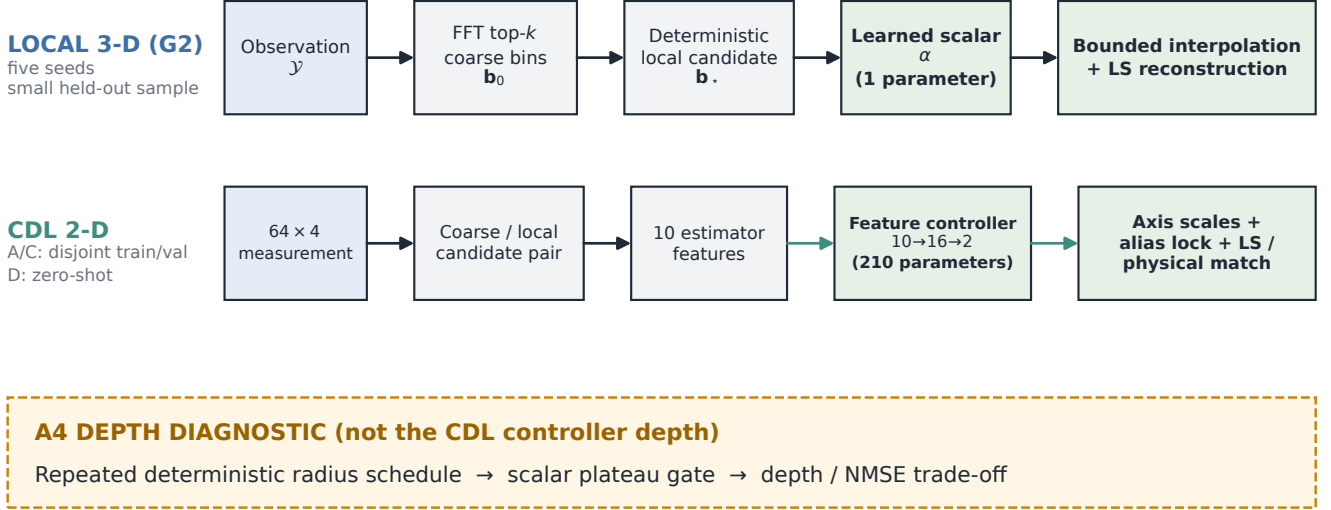


Fig. 2. Implemented refinement paths and evidence scope. The local three-dimensional experiment learns one scalar interpolation parameter. The CDL experiment uses a separate 10–16–2 feature controller. A4 repeats deterministic local searches and applies a scalar plateau gate; it is a depth diagnostic, not the depth of the CDL controller.

Consequently, the G2 result tests whether a learned global step length improves a deterministic local candidate; it does not test a deep stack of trainable Tensor-OMP layers.

C. Feature-Conditioned CDL Controller

The CDL path uses a separate controller because a single shared delay–angle scale produced strong axis coupling. A standardized feature vector $\mathbf{z} \in \mathbb{R}^{10}$ records support density, selected spectral energy, grid residual energy, delay/angle displacement statistics, support uniqueness, and design-matrix conditioning. The controller is

$$\begin{bmatrix} \alpha_\tau \\ \alpha_\theta \end{bmatrix} = 1.2 \sigma(\mathbf{W}_2 \tanh(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1) + \mathbf{b}_2), \quad (7)$$

with dimensions 10 → 16 → 2, for 210 trainable parameters. At the four-element ULA Nyquist bin, the angle coordinate is locked to its coarse value to avoid changing a valid endfire representation through an aliased continuous update.

For each training sample, the current predicted set is matched to the reference set with Hungarian assignment. The optimized objective is

$$\mathcal{L} = \text{NMSE}(\hat{\mathbf{y}}, \mathbf{y}_{\text{clean}}) + \lambda_p [\text{SL}_{10.5}(\hat{\boldsymbol{\tau}}, \boldsymbol{\tau}) + \text{SL}_{10.25}(\mathbf{e}_\theta/30)], \quad (8)$$

where $\mathbf{e}_\theta$ is the minimum valid projected-angle error over the ULA alias candidates. Assignment and the alias-aware angle calculation are recomputed from the current prediction. This makes supervision permutation-invariant while keeping

the physical objective aligned with test-time metrics.

D. Separate Repeated-Refinement Diagnostic

A4 studies a different executable path. Starting from the same grid support, it repeats deterministic local searches with radii $\{0.45, 0.28, 0.18, 0.11, 0.07, 0.045, 0.03, 0.02\}$ and recomputes LS at each depth up to $K = 8$. If q_k is NMSE in dB after depth k , the plateau gate stops at

$$\hat{K} = \min\{k \geq K_{\min} : q_{k-1} - q_k < \gamma\}, \quad (9)$$

or at K if the set is empty. This path has no trained CDL feature controller and should not be read as network depth. Its evidence supports only a high-load, 20 dB computation-quality trade-off under the tested schedule.

V. CONDITIONAL ANALYSIS AND VALIDATION BOUNDARIES

The analysis below explains the local mechanism and the validation checks; it is not a global recovery or efficiency theorem. Its assumptions are separable differentiable atoms, a correct selected support for the local mismatch argument, and local differentiability within the bounded refinement neighborhood.

A. Support-Conditioned Mismatch

Let $\mathbf{a}(\mathbf{b})$ be a differentiable tensor atom and suppose the true component lies at $\mathbf{b}_0 + \delta$ near a selected grid point $\mathbf{b}_0$. Locally,

$$\mathbf{a}(\mathbf{b}_0 + \delta) = \mathbf{a}(\mathbf{b}_0) + \mathbf{J}(\mathbf{b}_0)\delta + \mathcal{O}(\|\delta\|_2^2), \quad (10)$$

where $\mathbf{J}$ is the atom Jacobian. If local search returns an offset candidate δ_* , the first-order mismatch after bounded interpolation is proportional to

$$\|\mathbf{J}(\mathbf{b}_0)(\delta - \alpha \odot \delta_*)\|_2. \quad (11)$$

Learning the interpolation scale can therefore reduce quantization mismatch when the selected support and local candidate are informative. Equations (10)–(11) do not guarantee improvement after a wrong support decision; the bound in (4) only limits how far such an error can propagate.

B. Depth-Quality Diagnostic

For the repeated-refinement path, define the relative depth saving as

$$s_K = 100 \left(1 - \frac{\hat{K}}{K} \right) \%, \quad \Delta q = q_{\hat{K}} - q_K. \quad (12)$$

The gate is useful only when s_K is positive and the quality gap Δq stays within a chosen tolerance. The current A4 study uses this operational criterion; it does not establish that (9) is optimal. Its failure across the SNR sweep further shows that small one-step improvement is not a sufficient SNR-invariant stopping statistic for the tested curves.

C. CRLB Implementation Check

For complex Gaussian observations with mean $\mu(\eta)$ and covariance $\sigma^2 \mathbf{I}$, the real-parameter Fisher information used in the audit is

$$[\mathbf{F}(\eta)]_{ij} = \frac{2}{\sigma^2} \operatorname{Re} \left\{ \left(\frac{\partial \mu}{\partial \eta_i} \right)^H \frac{\partial \mu}{\partial \eta_j} \right\}. \quad (13)$$

Analytic derivatives agree with finite differences to relative error 5.73×10^{-10} in the recorded audit. In the single-target pilot, the mean 30 dB RMSE-to-CRLB ratio is 1.0067 and the mean slope error is 0.0067. These values validate the derivative implementation and high-SNR trend only; they do not establish multi-target CRLB tightness for the learned estimators.

VI. EXPERIMENTS AND ANALYSIS

The experiments ask three separate questions: whether a learned scalar improves a deterministic local candidate, whether a frozen feature controller improves channel and physical parameters under standards-aligned CDL channels, and whether deterministic repeated refinement can be shortened without an unrestricted adaptive-depth claim.

A. Evaluation Protocol

The local experiments use tensor shape $128 \times 16 \times 32$. G2 evaluates $L \in \{2, 4, 8\}$ at SNR = 20 dB against grid Tensor-OMP on matched held-out samples, using five seeds with two training and 30 test scenes per L and seed. The matched continuous-recovery campaign uses the same shape, $L \in \{2, 4, 8\}$, SNR $\in \{0, 10, 20, 30\}$ dB, five seeds, and 20 paired scenes per cell. The CDL experiment uses 64 subcarriers and a four-element ULA over the same SNR set and $L \in \{4, 8, 16\}$. The controller is trained on CDL-A/C with separate validation seeds and frozen before evaluation on five new seeds with 50 samples per profile and seed, for 6000 held-out A/C sample rows. CDL-D is reserved for zero-shot profile-shift evaluation.

The primary reconstruction metric is measurement-domain NMSE in dB. For A4, depth is used as the FLOPs proxy because each deterministic refinement step performs the same local-search kernel. A cell is treated as positive only when the mean depth saving reaches the planned 25% level and the NMSE gap to fixed $K = 8$ remains within the 0.5 dB tolerance.

B. Main Result: Bounded Off-Grid Refinement

Figure 3a shows that the learned scalar improves over grid Tensor-OMP across all evaluated target counts and seeds. The mean L -cell gain is 5.8956 dB, the smallest L -cell gain is 5.3863 dB, and the seed-mean 95% CI is [5.7830, 6.0083] dB. The 30 scenes per L and seed strengthen the population estimate while preserving the local synthetic scope.

This result supports the central local mechanism claim. Standards-aligned external evidence is reported separately below.

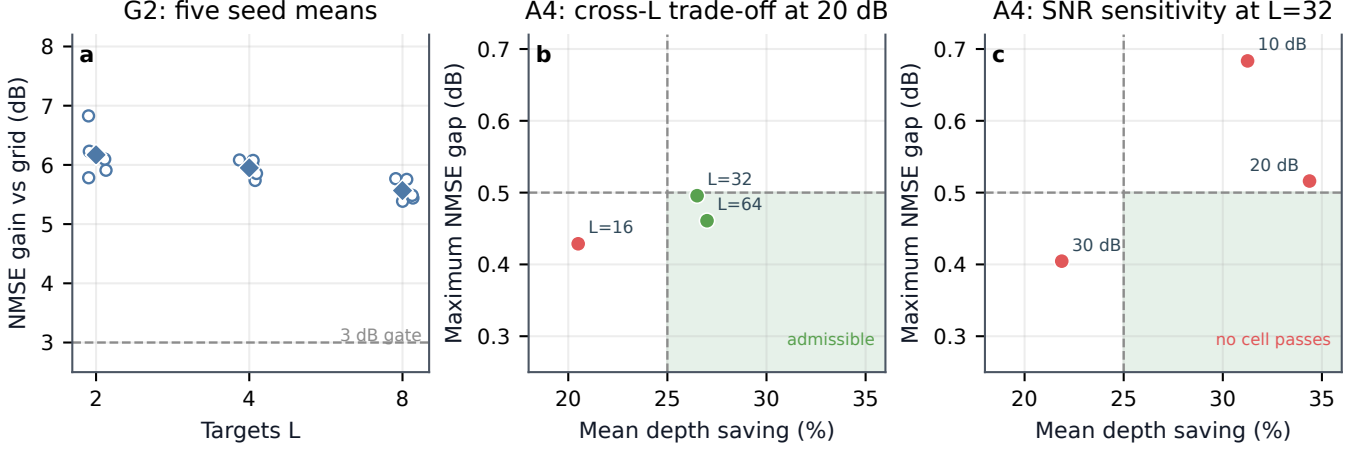


Fig. 3. Local evidence and depth boundary. (a) Open circles show the five seed means for each target count; diamonds show their means. All cells exceed the 3 dB gain gate. (b) Cross- L depth saving versus the maximum NMSE gap at 20 dB; the shaded region marks the prespecified $\geq 25\%$ saving and ≤ 0.5 dB gap criteria. (c) The same trade-off across SNR at $L = 32$; no SNR cell satisfies both criteria.

C. Matched Continuous-Recovery Comparison

The matched campaign compares grid, deterministic local refinement, the frozen source-trained scalar, a known-order 3-D NOMP-inspired path, and fixed-sweep PARAFAC-ALS on 1200 paired scenes. Figure 4 makes the intended conclusion explicit: NOMP-inspired refinement is the accuracy-oriented method, with mean NMSE -53.390 dB and a 42.1756 dB mean advantage over the learned scalar, while requiring an $8.106\times$ larger median runtime. The learned scalar is therefore a bounded amortized alternative, not an accuracy winner. PARAFAC-ALS reaches -16.855 dB mean NMSE; separable FB-ESPRIT remains timing-only because its marginal axis estimates lack cross-axis component pairing.

This comparison assumes known target count, dense tensor measurements, and the matched sinusoidal generator. It is a 3-D mechanism adaptation rather than a reproduction of sparse-resource NOMP-OFDM-ISAC with CFAR stopping. Table II retains the full aggregate values and the ESPRIT availability boundary.

D. Standards-Aligned CDL Evaluation

On held-out CDL-A/C seeds, the frozen controller improves all three aggregate metrics over grid Tensor-OMP. Across five seeds and 50 samples per profile and seed, the channel-NMSE gain is 6.2286 dB (95% seed-level CI $[6.1018, 6.3553]$), the physical-delay RMSE reduction is 10.1241 ns ($[8.6931, 11.5551]$), and the projected broadside-angle RMSE reduction is 0.5277° ($[0.3122, 0.7432]$).

The positive statement is aggregate and profile-scoped. At $L = 16$, mean delay reduction is -0.8112 ns, and some individual cells have negative angle deltas. Zero-shot CDL-D improves projected angle but degrades channel NMSE and delay, preventing a profile-universal claim.

E. Efficiency Trade-Off and SNR Boundary

Figure 3b evaluates the plateau gate over $L \in \{16, 32, 64\}$ at SNR = 20 dB. The high-load cells, $L = 32$ and $L = 64$,

satisfy the narrowed early-stop criterion: the minimum mean saving over $L \geq 32$ is 26.5000% , and the maximum seed-cell mean NMSE gap is 0.4957 dB. $L = 16$ is boundary evidence because its mean saving is only 20.5000% . This diagnostic uses five seeds with two training and five test curves per cell; it strengthens the distributional evidence but remains a local computation-quality mechanism rather than a deployment guarantee.

Figure 3c shows the boundary of the early-stop result; the complete sensitivity table remains in the supplementary display artifacts. At $L = 32$ over SNR $\in \{10, 20, 30\}$ dB, the scalar plateau gate is inconclusive as a mechanism across SNR. The minimum SNR-cell mean saving drops to 21.8750% , below the 25% target, while the maximum SNR-cell gap reaches 0.6834 dB, above the 0.5 dB tolerance.

A follow-up threshold-frontier diagnostic tests whether this failure is only a coarse-calibration artifact. It compares the train-selected scalar rule with a guarded train rule and a test-oracle threshold frontier. The result remains negative for a broad scalar-threshold claim: the standard train-selected policy has a minimum SNR mean saving of 15.6250% and a maximum SNR mean gap of 0.6691 dB, while even the diagnostic test-oracle frontier has a minimum SNR mean saving of 0.0000% . This separates a useful 20 dB high-load trade-off from a general SNR-axis depth policy. A stronger learned gate with explicit SNR or uncertainty features would be needed before A4 could be presented as stable across SNR.

VII. LIMITATIONS AND FUTURE WORK

The reported results combine controlled synthetic/off-grid evidence with standards-aligned held-out CDL-A/C evaluation. They do not constitute instrumented-receiver or profile-universal validation of an ISAC receiver. The one-parameter local model, the 210-parameter CDL controller, and the deterministic A4 depth diagnostic are separate implementations; a unified multi-stage learned estimator remains unvalidated.

The first limitation is external validity across channel families. The trained controller improves aggregate channel, delay,

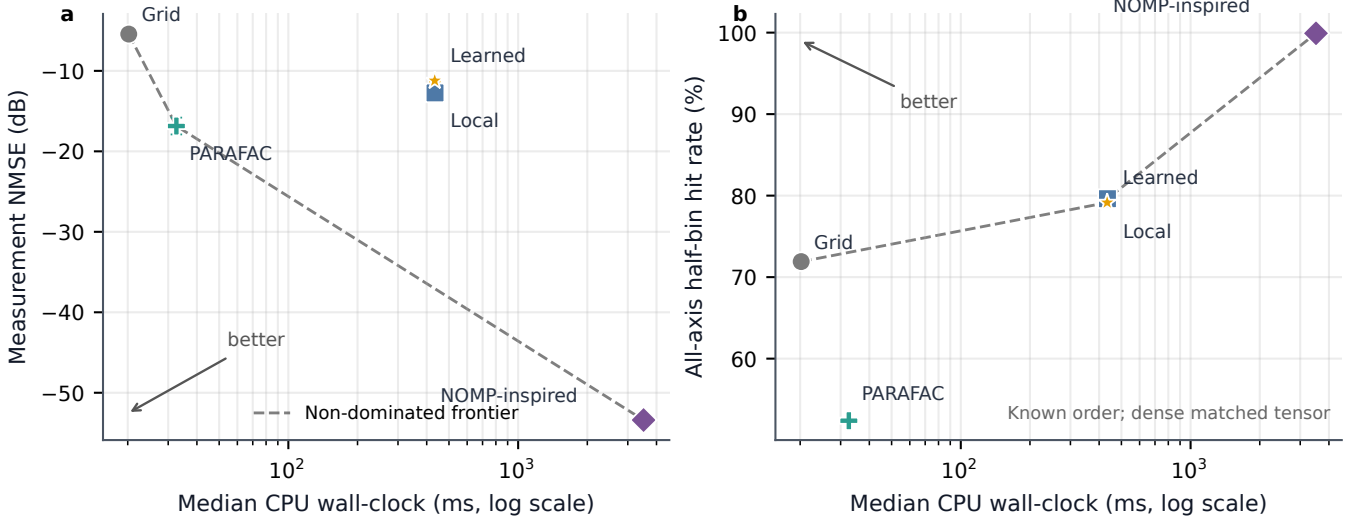


Fig. 4. Matched compute-accuracy evidence over five seeds, four SNRs, three target counts, and 20 scenes per cell. (a) Measurement NMSE with seed-mean 95% confidence intervals. (b) All-axis half-bin hit rate. Dashed lines connect non-dominated methods. The learned scalar is cheaper than NOMP but is not Pareto-optimal under end-to-end wall-clock; the NOMP-inspired path dominates matched-synthetic accuracy at higher runtime.

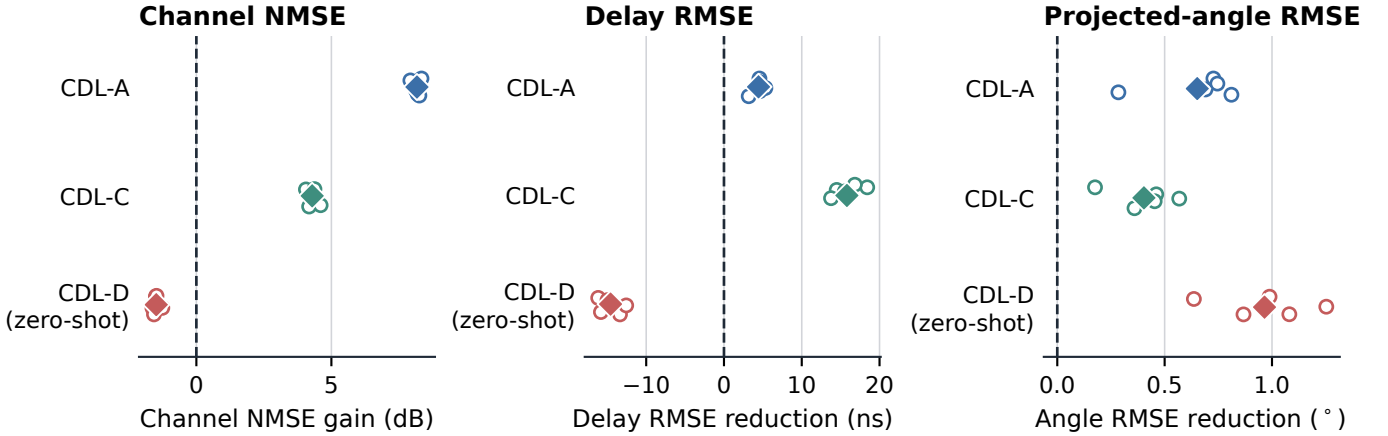


Fig. 5. Scaled CDL physical evaluation versus grid Tensor-OMP. Open circles are seed-level means over SNR and target count; diamonds are profile means. CDL-A/C improve channel NMSE by 8.1690/4.2881 dB, delay RMSE by 4.4421/15.8061 ns, and projected-angle RMSE by 0.6518°/0.4036°. Zero-shot CDL-D degrades channel NMSE and delay despite improving projected angle.

and projected-angle metrics on held-out CDL-A/C seeds, but zero-shot CDL-D degrades channel NMSE and delay RMSE. The result is therefore profile-scoped. DeepMIMO remains limited to scalar-transfer proxies without a matched trained physical evaluator, and hardware-in-the-loop evidence is absent.

The second limitation is the scope of adaptive depth. The positive early-stop evidence is restricted to high-load settings at the core 20 dB SNR condition, with the strongest main-text support for $L \geq 32$. The lower-load $L = 16$ cell is a boundary case. The SNR sweep at $L = 32$ is also mixed: the minimum SNR-cell mean saving is 21.8750%, and the maximum NMSE gap reaches 0.6834 dB. A threshold-frontier diagnostic reinforces this boundary: finer scalar-threshold selection does not recover a stable SNR-axis policy, and the diagnostic test-oracle frontier still has a minimum SNR mean saving of 0.0000%. A subsequent SNR/curve-feature gate diagnostic also fails to

rescue the claim: the lightweight kNN gate has a minimum SNR mean saving of 21.8750% and a maximum SNR gap of 0.6558 dB, while the per-curve quality oracle still falls to 21.8750% minimum SNR mean saving. Thus, the plateau gate is best interpreted as a useful high-load efficiency mechanism under the tested core condition, not as a depth policy that is stable across SNR.

The third limitation is robustness to hardware impairments. The combined impairment stress test does not yet support a main robustness claim. It is still useful as failure-mode evidence, because it shows where the current training and refinement recipe is brittle, but a stronger contribution would require robust training or explicit impairment-aware calibration.

The fourth limitation is comparator scope. G2 now uses 30 held-out scenes per target count and seed, and the matched campaign uses 20 paired scenes in every SNR-target-count-

seed cell. The NOMP-inspired comparator nevertheless assumes known target order, a dense tensor, and the matched sinusoidal generator; it is not the sparse-resource CFAR formulation of NOMP-OFDM-ISAC. Separable forward-backward ESPRIT remains timing-only because the current implementation does not pair components across axes. The resulting Pareto plot therefore establishes a matched-regime compute-accuracy trade-off, not universal dominance.

The fifth limitation is theoretical coverage. The CRLB analysis checks analytic derivative consistency and a single-target high-SNR trend. It does not establish tightness for the full multi-target asymptotic regime, nor does it remove the support-conditioning assumption behind the local off-grid argument.

Future work should therefore integrate the learned paths into one executable estimator, add matched-accuracy continuous-recovery comparators, enlarge the local statistical study, close the CDL-D and DeepMIMO profile-shift gap, and train against hardware impairments.

VIII. CONCLUSION

This work studies learned bounded refinement after TensorOMP support selection for mmWave MIMO-OFDM ISAC. The central idea is to interpolate between an interpretable grid support and a deterministic local candidate, then recompute the least-squares reconstruction. The evaluated learned paths are a one-parameter local scalar and a separate feature-conditioned CDL controller; the repeated-refinement depth study is a deterministic diagnostic rather than an integrated network.

The evidence supports a narrowed but coherent claim. Local synthetic evaluation shows consistent gains for $L \in \{2, 4, 8\}$. In scaled held-out CDL-A/C evaluation, the frozen controller improves channel NMSE by 6.2286 dB, delay RMSE by 10.1241 ns, and projected-angle RMSE by 0.5277° on average. The high-load early-stop result remains restricted to the core SNR condition.

The limitations are part of the contribution boundary. Zero-shot CDL-D degrades channel and delay metrics despite improving projected angle, so the external result is not profile-universal. Trained DeepMIMO physical evaluation, instrumented-receiver validation, robust hardware-impairment training, unrestricted adaptive-depth optimality, and full multi-target CRLB tightness remain open.

The result is strongest when framed as interpretable, claim-bounded post-support learning rather than as a completed deep-unfolding system. Within that scope, bounded interpolation improves local sparse tensor recovery and held-out CDL-A/C metrics while preserving enough structure to expose profile shift, sample-size limits, and the failure of the tested stopping statistic across SNR.

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APPENDIX A EXPERIMENT TABLES

The per- L G2/A4 table, full A4 SNR sensitivity, scalar-threshold frontier, and SNR-aware gate tables are retained as generated supplementary artifacts in the manuscript display workspace. They are omitted here because Fig. 3 already carries the main local comparison and the three SNR tables are boundary diagnostics.

APPENDIX B REPRODUCIBILITY AND EVIDENCE MAP

The manuscript is anchored to the generated v2.3R evidence table and run manifests. In addition to the local G2, A4, Kruskal, CRLB, and impairment results, the evidence map includes the scaled trained CDL-A/C result, the matching-scale zero-shot CDL-D boundary, and DeepMIMO scalar-transfer proxies.

APPENDIX C DATA AND CODE AVAILABILITY

All synthetic generators, evaluation scripts, run manifests, aggregate tables, and figure builders needed to reproduce the reported results are available at <https://github.com/Adonis-0311/OFDM-MIMO-Project>. The repository preserves executable paths and durable result manifests for each paper-facing claim.

Paper-facing displays are generated by scripts in the manuscript workspace. The E4 benchmark adds executable grid, bounded-refinement, Torch-layer, separable forward-backward ESPRIT, and complex PARAFAC-ALS rows; its arithmetic counts are analytical proxies, and it does not assign estimator-accuracy claims to the comparator rows. The bibliography remains under active frontier expansion and metadata review.

The implementation is covered by 24 automated tests, including bounded refinement, permutation matching, controller gradients, continuous physical conversion, the Nyquist alias lock, separable ESPRIT frequency recovery, and rank-one complex PARAFAC reconstruction. Experiment commands, configurations, seeds, environment snapshots, and metrics are stored in run manifests beside each output.

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TABLE II

PAPER-SCALE MATCHED ACCURACY AND RUNTIME AT TENSOR SHAPE $128 \times 16 \times 32$. NMSE INTERVALS ARE 95% CONFIDENCE INTERVALS OVER FIVE SEED MEANS.

Method	NMSE (dB)	Median ms	Half-bin fraction	Availability
Grid FFT top- k + LS	-5.430 [-5.527, -5.333]	20.20	0.719	matched
Deterministic local refinement	-12.735 [-12.814, -12.655]	434.79	0.796	matched
Learned scalar refinement	-11.215 [-11.340, -11.089]	433.48	0.791	matched
NOMP-inspired refinement	-53.390 [-53.472, -53.308]	3513.99	0.999	matched
Complex PARAFAC-ALS	-16.855 [-17.900, -15.810]	32.57	0.524	matched
Separable FB-ESPRIT	-	see Table III	-	timing-only; no axis pairing

TABLE III

EXECUTABLE E4 COMPLEXITY COMPARISON AT TENSOR SHAPE $128 \times 16 \times 32$ AND $L = 8$. OPERATION COUNTS ARE ANALYTICAL PROXIES; TIMINGS ARE MEDIAN CPU WALL-CLOCK OVER FIVE REPEATS.

Method	Parameters	Operation proxy	Time (ms)	Scope boundary
Grid FFT top- k	0	5.24×10^6	7.42	on-grid support selection
Bounded refinement + LS	0	5.63×10^8	832.34	deterministic five-point local search
Torch alpha forward	1	3.77×10^7	10.41	prepared-bin forward path
Separable FB-ESPRIT	0	1.07×10^8	34.44	no cross-axis pairing; not full Unitary ESPRIT
Complex PARAFAC-ALS	0	6.34×10^7	41.62	five fixed ALS sweeps

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